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Leveraging related health phenotypes for polygenic prediction of impulsive choice, impulsive action, and impulsive personality traits in 1534 European ancestry community adults

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Abstract

Impulsivity refers to a number of conceptually related phenotypes reflecting self-regulatory capacity that are considered promising endophenotypes for mental and physical health. Measures of impulsivity can be broadly grouped into three domains, namely, impulsive choice, impulsive action, and impulsive personality traits. In a community-based sample of ancestral Europeans ($n = 1534$), we conducted genome-wide association studies (GWASs) of impulsive choice (delay discounting), impulsive action (behavioral inhibition), and impulsive personality traits (UPPS-P), and evaluated 11 polygenic risk scores (PRSs) of phenotypes previously linked to self-regulation. Although there were no individual genome-wide significant hits, the neuroticism PRS was positively associated with negative urgency (adjusted $R^2 = 1.61\%$, $p = 3.6 \times 10^{-7}$) and the educational attainment PRS was inversely associated with delay discounting (adjusted $R^2 = 1.68\%$, $p = 2.2 \times 10^{-7}$). There was also evidence implicating PRSs of attention-deficit/hyperactivity disorder, externalizing, risk-taking, smoking cessation, smoking initiation, and body mass index with one or more impulsivity phenotypes (adjusted R^2s: 0.35%–1.07%; FDR adjusted $ps = 0.05$–0.0006). These significant associations between PRSs and impulsivity phenotypes are consistent with established genetic correlations. The combined PRS explained 0.91%–2.46% of the phenotypic variance for individual impulsivity measures, corresponding to 8.7%–32.5% of their reported single-nucleotide polymorphism (SNP)-based heritability, suggesting a non-negligible portion of the SNP-based heritability can be recovered by PRSs. These results support the predictive validity and utility of PRSs, even derived from related phenotypes, to inform the genetics of impulsivity phenotypes.

KEYWORDS
delay discounting, educational attainment, genetic association, genome-wide association study, impulsivity, polygenic risk score, UPPS

1 | INTRODUCTION

Impulsivity is a multi-faceted psychological construct reflecting different aspects of self-regulation. Diverse measures of impulsivity have been consistently linked to psychiatric disorders and other adverse health conditions, such as obesity.[1–3] Individual measures of impulsivity are conceptually related but often quantitatively distinct. For example, one investigation of the latent structure of numerous impulsivity measures found three distinct domains.[4] These include *impulsive choice*, measured by delayed reward discounting tasks to determine the preferences for smaller immediate rewards over larger delayed rewards[5]; *impulsive action*, measured by the ability to inhibit a prepotent response, for example, through a Go/NoGo task[6]; and *impulsive personality traits*, assessed through self-reported questionnaires, such as the UPPS-P Impulsive Behavior Scales[7] or Barrett Impulsiveness Scales. In the case of impulsive personality traits, further fractionation of self-reported impulsivity is present via subscales. This tripartite division is probably at least partially based on notable differences in measurement, ranging from questionnaires to different types of behavioral tasks. Behavioral studies indicate these indicators to be generally stable and trait-like.[8–10]

From a genomic perspective, impulsive choice and personality traits have been the most extensively studied measures. So far, there have been two reported GWASs on delayed reward discounting. The first included 23,217 ancestral Europeans[11] who were members of personalized genomics company 23andMe. The second was conducted in a smaller but homogenous sample of 986 healthy substance-free ancestral European young adults.[12] However, these studies yielded only two genome-wide significant associations: rs6528024 (*GPM6B*) on X chromosome[11] and rs13395777 on chromosome 2.[12] Owing to their differences in sample size and data analytics, the two GWASs did not converge in their findings of individual variants and genes. Nevertheless, the larger study was sufficiently powered to establish the genomic basis of delay discounting with reported chip-based heritability of 12.2% and evidence of genetic overlap with other related phenotypes,[11] including neuropsychiatric conditions, smoking, body mass index, personality and cognitive measures, such as years of education.

GWASs were also conducted on the UPPS-P impulsive behavior scales in these same samples.[13,14] The smaller sample did not show any genome-wide findings.[14] However, the larger 23andMe sample[13] ($n = 22{,}861$) identified significant variants and established modest single-nucleotide polymorphism (SNP)-based heritability (5%–10%) for UPPS scales. In addition, there were also wide genetic overlap between impulsive personality traits and other impulsive, substance use, and neuropsychiatric traits, although the genetic correlations with substance use were more pronounced.[13] An update to this study with a larger sample size ($n > 123{,}509$) has confirmed the association with variants in Cell Adhesion Molecule 2 gene (*CADM2*) and identified additional promising genes with relevant neurobiological functions.[15] Subsequently, a recent PheWAS of *CADM2* in the UK Biobank[16] found significant associations with traits including substance use, risk-taking, health behavior, sleep, and dietary traits, some of which were also replicated in an independent sample exclusive to the updated GWASs,[15] providing further evidence for its relevance to impulsivity. Finally, a recent *CADM2* PheWAS in a number of different samples also found significant associations with various risky behaviors and measures of self-control,[17] further implicating this gene.

Genetic research is limited with regard to Impulsive action and only one GWAS has been conducted[18] in the aforementioned sample of 986 healthy young adults.[12,14] This study did not yield any significant SNP-level or gene-level results. Deficits in behavioral inhibition are a common feature of attention-deficit hyperactivity disorder (ADHD) and there have been more favorable genetic findings in relation to that condition,[19,20] which may ultimately relate to impulsive action and possibly the other two impulsivity domains.[21]

In contrast to self-regulatory traits, existing GWASs of related phenotypes, such as body mass index (BMI),[22] substance use,[23,24] and education,[25,26] have employed sample sizes in the millions through large biobanks and efforts from various genetic consortia. As a result, these GWASs are increasingly generating robust genetic signals at the individual variant level as well as in the form of a genome-wide polygenic risk score (PRS), which combines association signals into an aggregated variable additively. As compared with individual genetic variants, PRSs provide another potentially powerful tool to characterize the genetic basis of impulsivity phenotypes by leveraging their established SNP-based heritability and genetic correlation with related traits.[11,13,15] However, no studies have investigated the implications of these estimated heritability and genetic correlations for out-of-sample prediction.

The rationale for the current study was to use advances in PRS methodology and findings to inform the understanding of the genomic basis of self-regulatory traits. Here, we examined the empirical prediction of impulsive choice, impulsive action, and impulsive personality trait using polygenic scores of related traits, which was only previously implied by the established SNP-heritability and genetic correlations. Specifically, we focused on using PRSs of several related traits to contextualize previous heritability estimates of impulsivity and delineate the sources of genetic contribution to impulsivity variability. Extending previous recent phenotypic studies, we first examined phenotypic interrelationships among the various measures of impulsivity. Second, we conducted GWASs on measures of impulsivity, including response inhibition as a separate measurement domain of impulsivity in addition to the more commonly studied delay discounting and UPPS-P phenotypes, and compared association results of the most plausible risk variants based on existing large GWASs. Third, we tested associations between impulsivity and PRSs of phenotypes previously linked to impulsivity, such as alcohol use, educational attainment, tobacco use, and obesity. Collectively, the novel contributions of the current study were intended to substantively extend the limited existing literature on the genomic basis for self-regulatory traits.

2 | METHOD

2.1 | Study sample and data quality control

Participants were from an existing research registry at St. Joseph's Healthcare Hamilton in which community adults were invited to

TABLE 1 Descriptive statistics for the genetic subsample ($n = 1534$).

Category	Variable	Median	Mean (SD)
Demographics	Age	28	33.91 (14.04)
	Sex (% males)	N/A	43%
	Income level	6	5.47 (2.79)
	Years of education	15	15.07 (2.83)
Anthropometric measures	Weight (kg)	79.4	81.38 (19.22)
	BMI (kg/m^2)	26.54	27.85 (7.82)
Substance use	Drinks per week	6	8.86 (12.02)
	Cigarettes per day	0	0.69 (1.3)
Impulsive actions	NoGo trial accuracy	0.67	0.66 (0.21)
	Go trial accuracy	1	0.99 (0.02)
Impulsive choice	lg10 small *k*-value	−1.59	−1.79 (0.78)
	lg10 medium *k*-value	−2.01	−1.96 (0.83)
	lg10 large *k*-value	−2.01	−2.11 (0.81)
Impulsive personality	UPPS negative urgency	2.25	2.19 (0.74)
	UPPS positive urgency	1.5	1.71 (0.63)
	UPPS sensation seeking	2.5	2.5 (0.75)
	UPPS lack of premeditation	1.75	1.7 (0.53)
	UPPS lack of perseverance	1.75	1.75 (0.48)

participate in an in-person assessment including DNA collection. The general eligibility criteria were minimal, primarily requiring participants to be legal adults interested in participating in research studies and no contraindicating health conditions. Two thousand and ninety-eight participants were genotyped on Illumina Infinium® Global Screening Array as a membership service from 23andMe Inc; imputation was performed using the Michigan Imputation Server.[27] Standard quality control filters were applied to retain unique SNPs with INFO score >0.7, minor allele frequency (MAF) >0.05, missing rate <0.01, and Hardy–Weinberg disequilibrium p value $>1 \times 10^{-5}$ (5,654,384 SNPs). We restricted the analysis to unrelated individuals by retaining at most one member from each pair of samples with kinship coefficient greater than 0.0884. A final set of 1534 unrelated, genetically-confirmed European samples were included in the analyses using the subset of continental Europeans from 1000 Genomes Project as a reference (Figure S1). Data inclusion and exclusion criteria are summarized in a consort diagram (Figure S2) and further details on data processing can be found in Supplementary Materials. The demographic characteristics of the genetic subsample is described in Table 1.

2.2 | Phenotypic assessment

Impulsive choice was measured by delayed reward discounting using the Monetary Choice Questionnaire[28] (MCQ), giving three monetary reward magnitudes (small, medium, and large) of discounting rates (or *k*-values). The assessment comprises of dichotomous choices between small immediately available rewards and larger delayed rewards, with the items preconfigured to specific *k* values. Choice patterns can in turn be inferred as reflecting a certain discounting preference. Three additional questions were added to detect poor responses: a preference for a larger amount over a smaller amount at no delay is considered the correct response, and those with low effort/attention (i.e., 2/3 questions incorrect) were excluded ($n = 9$). Delay discounting has been found to have robust temporal reliability.[9,29] The continuous delay discounting phenotypes, were processed using a standard pipeline[30] at large, medium and small rewards and were log10-transformed.

Impulsive action was measured by trial accuracies from a Go/No-Go task. The "Go" stimulus requiring an emitted response was an "X" (85% of stimuli, 68 trials) and the "NoGo" stimulus requiring an inhibited response was a "K" (15%, 12 trials). The primary index of behavioral inhibition was errors of commission (i.e., providing a positive Go behavior in response to a NoGo stimulus), but with consideration of errors of omission also (i.e., failure to respond to a Go trial). Both were standardized to give an error rate between 0 and 1. For quality control, we removed those with a commission error rate greater than 0.8 and those with omission error accuracy below 0.5, suggesting categorically low task effort.[31] Performance on impulsive action tasks has been found to be generally stable over time in the absence of direct interventions.[32]

Impulsive personality traits were measured by the S-UPPS-P scale,[7] which consists of 20 items and was used to construct the five factors of impulsive personality: Positive Urgency (PU), Negative Urgency (NU), Lack of Perseverance (PER), Lack of Premeditation (PM), and Sensation Seeking (SS), with good internal reliability (standardized Cronbach's Alpha: NU = 0.79; PU = 0.80; SS = 0.72; PRE = 0.8; PERS = 0.7).

2.3 | Phenotypic structural analyses

Paralleling earlier work,[4] the latent phenotypic structure of impulsivity was examined using four successive structural equation models (SEMs): (1) a unidimensional model with all indicators; (2) a two-factor model with task-based indicators in one factor and impulsive personality traits in the second; (3) a three-factor model reflecting impulsive choice, impulsive action, and impulsive personality traits; and (4) a modified three-factor model, removing SS from impulsive personality traits. Model 4 was based on the previous final best-fitting model and the perspective that sensation seeking reflects variation in the preferred level of arousal/stimulation, rather than a self-regulatory trait.[4] Model 4 was hypothesized to exhibit the best fit. Since only one indicator of impulsive action was used, it was incorporated into the model as a manifest variable. Model fit was based on the confirmatory fit index (CFI, >0.90), Tucker Lewis Index (TLI, >0.90), Root Mean Square Error of Approximation (RMSEA, <0.08), standardized root mean square residual (SRMR, <0.08), and decreases in χ^2, AIC, and BIC.[33]

2.4 | Genome-wide association analyses

The response variables were the individual impulsivity measures or the latent impulsivity factor scores (per above). A linear regression model assuming additive genetic effects was used to examine the association between each SNP with the response variable while adjusting for sex, age, income genotyping array, and the first 10 genetic principal components (PCs). Genome-wide association of each impulsivity phenotype was conducted using PLINK2.[34] We examined the impulsivity association results of previously reported SNPs that are genome-wide significant for ADHD,[19] delay discounting,[11] drug experimentation, impulsive personality traits[13] for a trait-specific comparison. In light of the two consortium-sized GWASs currently under preprint, we extracted genome-wide significant SNPs for ADHD,[20] delay discounting,[11] and impulsivity personality traits,[15] yielding 50 genome-wide significant SNPs for which 43 are present in our samples (Table S1), including 42 genotyped or imputed SNPs and 1 proxy SNP in high linkage disequilibrium ($r^2 > 0.8$) with the lead SNP that was not available. To account for the SNPs specific to each impulsivity trait, the nominal significance for replication is based on false discovery rate (FDR) adjusted p-values being less than 0.05. We also expanded the search to the 561 genome-wide significant SNPs associated with externalizing[23] present in our data (Table S2). These established genome-wide significant SNPs were also highlighted in the Manhattan plots.

2.5 | Related health phenotype curation and PRS construction

Among phenotypes previously suggested to be genetically correlated with impulsivity,[11,13,15] we prioritized ADHD,[19] AUDIT total score,[35] smoking[36] (including cessation, initiation, cigarettes per day), cannabis use disorder (CUD),[37] externalizing,[23] neuroticism,[38] general risk tolerance,[39] body mass index (BMI),[40] and educational attainment (EA).[25] The rationale is based on selecting behavioral traits that are relevant to the non-clinical sample comprised of adults while controlling the type I error rate under multiple comparisons. In summary, we included three categories of the most commonly used psychoactive drugs, all of which have been linked to self-regulatory traits.[11,13] Other than substance use, the only other psychiatric condition included was ADHD, for its well-established phenotypic association with lack of impulse control.[21] These domains were complemented by phenotypes that do not directly map to a psychiatric diagnosis (e.g., EA and BMI) based on the evidence of previous linkage to impulsivity phenotypes.[11]

PRSs were constructed using lassosum,[41] which performs an elastic-net regression using summary statistics and a matching genomic reference panel for the target population (EUR.hg19). Model validation was based on within-sample phenotype that closest matches the external phenotype or a pseudo-validation when such phenotype was unavailable. For within-sample validation, the penalty parameters were selected based on maximizing the adjusted R^2 while adjusting for the first 10 genetic PCs, age, sex, income, and an indicator for genotyping array. For more details, see Supplementary Material. The most recent and the largest studies were used as sources of summary statistics when available (summarized in Table S3).

The performance of a PRS was evaluated by the prediction R^2 with respect to each impulsivity measure or latent impulsivity factor score using the same linear model adjusting for sex, age, income, genotyping array, and the first 10 genetic PCs. For each pair of PRS and impulsivity phenotype, the standardized regression coefficient (β) and the association p-value were used to inform the direction and significance of association, respectively. All analyses were conducted in Statistical Software R version 4.0.5.[42] We further assessed the combined predictive performance of all PRSs with respect to each impulsivity measure and domain, by calculating the difference in adjusted R^2 between the full model including the 11 PRSs, sex, age, income, genotyping array, and the first 10 genetic PCs, against the reduce model without any PRSs.

2.6 | Partitioning educational attainment PRS based on genes expressed in brain tissues

Integrating gene expression and GWAS data can show one or multiple tissues or cell type as the primary vehicle for genetic mechanisms to influence complex traits. For example, brain tissues have been implicated for EA[26,43] while both brain and adipose tissues have been identified to be relevant for BMI.[40] To understand the tissue specific contribution to the shared genetic signals between EA and impulsivity, we further partitioned the PRS[44] based on whether a SNP is in proximity of genes significantly expressed in brain tissues or not. We obtained the list of significantly expressed genes using the Genotype-Tissue Expression (GTEx) database,[45] version 8, only those significantly expressed in the brain tissue with an FDR adjusted p-value <0.05 were selected, giving a total of 29,131 autosomal genes. We filtered SNPs based on a distance of ±2000 bp to the start and the end position of each selected gene.

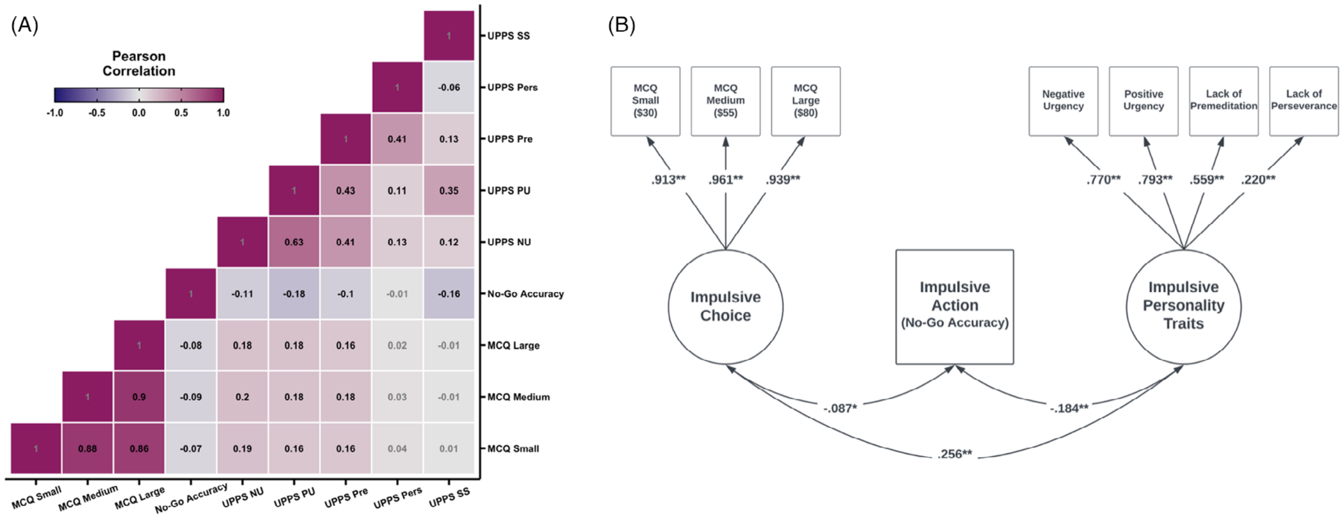

FIGURE 1 Phenotypic structure of impulsivity in 1534 community adults. (A) Zero-order associations among the impulsivity variables. (B) The optimal structural equation model of a tripartite approach comprising impulsive choice, impulsive action, and impulsive personality traits.

TABLE 2 Structural equation model fit indices.

	χ^2 (df)	CFI	TLI	RMSEA	SRMR	AIC	BIC
1 factor	1669.626 (27)	0.759	0.679	0.199	0.139	20144.35	20288.41
2 factor[a]	459.628 (26)	0.936	0.912	0.104	0.065	18936.35	19085.75
3 factor[b]	417.168 (25)	0.942	0.917	0.101	0.057	18895.888	19050.621
3 factor[c]	259.482 (18)	0.963	0.943	0.094	0.047	15547.053	15685.779

[a]Task versus questionnaire model (UPPS variables are one and delay discounting and No-Go accuracy are the second).
[b]Includes sensation seeking in the impulsive personality traits factor.
[c]Three-factor model with sensation seeking remove.

3 | RESULTS

3.1 | Phenotypic structure of impulsivity

As expected, the impulsivity indicators exhibited highly heterogeneous associations, ranging from very high overlap to effectively no association (Figure 1). In the SEMs, the results replicated the previously observed tripartite phenotypic structure (Table 2). Specifically, the unidimensional and two-factor models were inferior to the three-factor models and the three-factor model removing sensation seeking exhibited the best model fit. Very high factor loadings were present for delay discounting (>0.90), supporting consolidation for genomic analyses, whereas more variability was present among the impulsive personality traits, supporting individual trait-level genomic analyses.

3.2 | Genome-wide association analyses

There were no genome-wide significant associations between individual variants and any impulsivity traits nor the three factors (Figures S3–S8). None of the 43 genome-wide significant SNPs were significantly associated with their respective impulsivity traits after FDR control (Table S1), neither were any of the 561 genome-wide significant SNPs from the externalizing GWAS[23] (Table S2).

3.3 | Polygenic risk score associations

All constructed PRSs for the related traits were significantly associated with their corresponding trait or a proxy trait after FDR control (q-value <0.05; Table S4). Among them, the BMI, EA, smoking initiation, and AUDIT PRSs were the strongest predictors of their corresponding trait in the registry (Table S4), explaining 15.37%, 3.62%, 2.96%, and 1.17% of the trait variance adjusted for covariates, respectively. For ADHD, we used the WHO ADHD Part A dimensional item counts and this association was significant ($q = 0.03$) with the explained variance at 0.26%. CUD was significantly associated with the CUDIT severity subscale ($q = 0.02$) but not the CUDIT total score ($q = 0.12$). No focal indicators were available for risk tolerance or externalizing. Thus, we assessed the within-sample performance of the general risk tolerance PRS using the UPPS item "I quite enjoy taking risks" as the validating phenotype ($q = 0.01$), and while the externalizing PRS was associated with many of the externalizing phenotypes available in the registry, the strongest association was with cigarettes per day ($q = 1.1 \times 10^{-7}$). For the remaining smoking

TABLE 3 Significant associations between polygenic risk scores (PRSs) of related traits and impulsivity choice.

PRSs of related traits	logK small				logK medium				logK large			
	β	Adj R^2 (%)	p value	q value[a]	β	Adj R^2 (%)	p value	q value[a]	β	Adj R^2 (%)	p value	q value[a]
ADHD	0.09	0.79	0.0003	0.003	0.08	0.64	0.0009	0.009	0.08	0.6	0.001	0.01
Smoking	0.1	1.07	0.00003	0.0006	0.08	0.62	0.001	0.01	0.07	0.43	0.006	0.04
Externalizing	0.07	0.49	0.003	0.02	0.04	0.08	0.14	0.26	0.05	0.2	0.04	0.12
BMI	0.06	0.28	0.02	0.08	0.07	0.46	0.005	0.03	0.06	0.41	0.007	0.04
Education	−0.11	1.2	0.00001	0.0002	−0.12	1.52	7.65E-07	0.00003	−0.1	1.55	6.38E−07	0.00003

[a]False discovery rate adjusted p values (q values) were calculated based on multiple comparisons of 11 PRSs and 10 impulsivity measures.

phenotypes, PRSs of cigarettes per day and smoking cessation explained 0.59% ($q = 0.002$) and 0.78% ($q = 0.0005$) of the phenotypic variance of cigarettes per day, respectively, although reported smoking behavior in the sample was low.

The phenotypic structure results suggested consolidation of impulsive choice, but individual trait-level genomic analyses for impulsive personality trait. When examining the association between constructed PRSs and impulsivity choice, there were six significant associations in the expected direction, including ADHD, smoking initiation, smoking cessation, neuroticism, BMI and EA PRSs (Table S5). The strongest association was between the EA PRS and impulsive choice, with the EA PRS inversely associated with impulsive choice ($\beta = -0.13$) and explaining 1.68% ($q = 3.66 \times 10^{-6}$) of the phenotypic variance. The BMI PRS was associated—albeit weakly—in the expected direction ($\beta = 0.07$ in both cases) with impulsive action and choice (adjusted $R^2 = 0.36$–0.45%; $qs = 0.03$–0.02).

We focused on further examined the association between constructed PRSs and individual impulsivity phenotypes (Table S6). All PRSs, except AUDIT, smoking cessation, and cigarettes per day, were significantly associated with certain impulsivity indicators after FDR control at $q < 0.05$. There were no significant associations between indicators of impulsive action and any PRSs. ADHD, smoking initiation, and EA PRSs were significantly associated with all magnitudes of delay discounting (Table 3), however, there were variations among the three magnitudes of discounting preference. In particular, for ADHD and smoking initiation, the small log10-K values had the strongest signals of association. For EA, the large log10-K values was the most significantly associated magnitude. In contrast, the PRS of BMI was positively associated with the medium and large magnitudes, while the externalizing PRS was only positively associated with the small magnitude ($\beta = 0.07$; adjusted $R^2 = 0.49$; $q = 0.02$). The neuroticism PRS was significantly associated with higher levels of impulsive personality indicators, except Sensation Seeking (Table 4), and the stronger signals were with Positive, Negative Urgency and Lack of Premeditation (adjusted $R^2 = 0.65\%$–1.61%, $qs = 0.009$–0.00003). The externalizing PRS was weakly associated with a higher level of urgency for both the Positive and Negative indicators ($R^2 = 0.35\%$–0.36%, $qs = 0.05$). In addition, higher ADHD and EA PRSs were also significantly associated with higher and lower values of Lack of Premeditation, respectively, explaining 0.81% ($\beta = 0.09$; $q = 0.003$) and 0.59% ($\beta = -0.08$; $q = 0.01$) of the phenotypic variance. Finally, the general risk tolerance PRS was associated only with Sensation Seeking ($\beta = 0.1$; adjusted $R^2 = 0.85\%$; $q = 0.003$).

Combining the 11 PRSs improved the genetic predictive performance for both impulsive choice and individual impulsivity personality measures: the combined PRSs explained 2.2% ($q = 0.00001$; Table S7) of the delay discounting phenotypic variance and 0.91%–2.46% ($qs = 0.01$–5.5×10^{-6}) of the phenotypic variance for UPPS scales, excluding Lack of Perseverance. These corresponding to 8.7%–32.5% of their reported SNP-based heritability (Table S3), suggesting a non-negligible portion of the SNP-based heritability can be recovered by the combined PRSs of self-regulatory traits.

Finally, we observed consistency in the direction of PRS-impulsivity correlations with respect to the corresponding phenotype-impulsivity correlations for all nominally significantly associated PRS-impulsivity pairs (Figure S9). However, among those well-powered related trait PRSs (i.e., BMI, AUDIT, and EA), there were some discrepancies in the size of PRS-impulsivity correlations vs. phenotype-impulsivity correlations, particularly between impulsive personality and AUDIT, years of education; and Sensation Seeking and BMI (Figure S9). In these cases, the phenotypic correlation was moderate, but the PRS explained little to no variance in these impulsivity traits.

3.3.1 | Interrogating the link between the educational attainment polygenic risk score and delay discounting

The most significant predictor of delay discounting was the EA PRS and thus we focused on the 116,011 SNPs with non-zero weights contributing to this association. Roughly 56% of these SNPs (65,223) were within 2 kbp of genes significantly expressed in any brain tissue, they together explained 2.01% of phenotypic variance of EA ($p = 8.6 \times 10^{-9}$; Table S8), or equivalently 55.7% of the phenotypic variance of EA explained by all SNPs in the PRS (3.62%; $p = 4.6 \times 10^{-14}$). In line with the proportion of SNPs included, both the brain specific portion of the PRS and the non-brain specific PRS ($p = 5 \times 10^{-6}$) were significantly associated with EA. In contrast, only

TABLE 4 Significant associations between polygenic risk scores (PRSs) of related traits and impulsivity personality traits.

PRSs of related traits	Negative urgency				Positive urgency				Sensation seeking				Lack of premeditation				Lack of perseverance			
	β	Adj R^2 (%)	p value	q value[a]	β	Adj R^2 (%)	p value	q value[a]	β	Adj R^2 (%)	p value	q value[a]	β	Adj R^2 (%)	p value	q value[a]	β	Adj R^2 (%)	p value	q value[a]
ADHD	0.05	0.19	0.05	0.13	0.06	0.26	0.03	0.08	0.05	0.16	0.06	0.16	0.09	0.81	0.0002	0.003	0.02	−0.04	0.51	0.64
Externalizing	0.07	0.35	0.01	0.05	0.07	0.36	0.009	0.05	0.06	0.27	0.02	0.08	0.06	0.3	0.02	0.07	0.01	−0.06	0.70	0.79
Neuroticism	0.13	1.61	3.6E-07	0.00003	0.08	0.65	0.0009	0.009	−0.02	−0.04	0.50	0.64	0.11	1.23	8E-06	0.0002	0.07	0.42	0.006	0.04
Risk tolerance	0.01	−0.05	0.59	0.72	0.05	0.14	0.07	0.17	0.1	0.85	0.0002	0.003	0.04	0.1	0.11	0.23	0	−0.06	0.87	0.92
BMI	0.08	0.63	0.001	0.01	0.05	0.15	0.07	0.17	−0.01	−0.06	0.82	0.88	0	−0.07	1.00	1.00	−0.02	−0.02	0.43	0.58
Education	−0.03	0.06	0.17	0.30	−0.04	0.08	0.13	0.26	0.05	0.15	0.07	0.17	−0.08	0.59	0.002	0.01	0.01	−0.05	0.61	0.72

[a]False discovery rate adjusted p values (q values) were calculated based on multiple comparisons of 11 PRSs and 10 impulsivity measures.

the brain specific PRSs were significantly associated with all three magnitudes of delayed discounting preferences, explaining 0.78%–1.02% of the phenotypic variance of delay discounting.

4 | DISCUSSION

The current study examined the genetics of impulsivity phenotypes in community-based adults of European ancestry using two strategies, a SNP-level GWAS approach and a PRS-based approach. In the first case, we found no statistically significant association between individual genetic variants and any measures of impulsivity. In the second case, we observed notable associations between the genome-wide PRS of educational attainment (EA) and delay discounting, and between PRS of neuroticism and impulsive personality traits. In fact, the combined PRSs recovered 8.7%–32.5% of the reported SNP-based heritability[11,13] for delay discounting and the UPPS scales. The SNP-based heritability of impulsivity can likely be stratified to genetic effects that overlap with related traits and genetic effects that are specific to impulsivity. Using an impulsivity PRS as a covariate can clarify the source of genetic overlap, but this is not possible because of the unavailability of an impulsivity PRS. As a result, the captured genetic variance by combined PRSs in the current investigation represents only the genetic component of impulsivity that overlaps with related traits. Nevertheless, these results suggest that the predictive performance of PRSs, even derived from phenotypes genetically correlated with impulsivity, can meaningfully capture a substantial proportion of SNP-based heritability of impulsivity measures in a study of small to moderate sample size.

We observed theoretically consistent associations between the genetic components of related traits and relevant impulsivity phenotypes, such as those for lack of premeditation, delay discounting and ADHD,[46] delay discounting with respect to smoking initiation, externalizing, and EA,[11] as well as negative and positive urgency, and lack of premeditation with neuroticism.[13] In particular, the neuroticism PRS being positively associated with both urgency indicators is in line with previous research demonstrating their positive genetic correlations with an internalizing latent factor that combines depression, neuroticism, and subjective well-being.[47] Likewise, the positive association between risk tolerance PRS and sensation seeking is consistent with their phenotypic overlap, where the sensation seeking subscale contains an item "I quite enjoy taking risks" that assesses risk taking. These results also confirmed previous findings linking established BMI loci to delay discounting and negative urgency.[48] More importantly, the directions of phenotype–phenotype correlation are consistent with PRS-phenotype association for the significant impulsivity traits (Figure S9).

In some case, the constructed PRSs explained more variance for impulsivity traits than the corresponding trait itself, such as the ADHD PRS with lack of premeditation and delay discounting (Figure S10). This could be because of: (1) a mismatch between the PRS phenotype (e.g. a diagnosis in the external GWAS) and the available phenotype (e.g. self-reported symptom counts), leading to decreased power of association between the PRS and its corresponding trait; and (2) the sample being comprised of non-clinical, predominantly non-smoking participants, thus weakening the association signal between the some of the related traits and their PRSs, and finally, (3) the underlying genetic influence being on impulsivity, an endophenotype, were in fact stronger than those on related traits. The latter would be theoretically consistent with the impulsivity variables serving as risk mechanisms for the health outcomes, rather than pleiotropic expressions with common genetic underpinnings.

Genetic evidence of the involvement of brain gene expression in impulsivity traits has been limited, where the predicted expression of a single gene, *CDK3*, in the hippocampus region had been reported for delay discounting.[11] The partitioned PRS results point to genes significantly expressed in the brain to be a plausible source of shared heritability (Table S8) between educational attainment and self-regulatory health. These suggest that the genetic contribution to delay discounting through the brain could be much more widespread, though identification of specific genes remain challenging because of the limited sample size.

A collateral result of interest is further support for a tripartite latent model of impulsivity phenotypes.[4] This framework was initially tested in a sample of young adults within a narrow age range (mean age = 21.5) who were required to report verified low levels of substance use (with biochemical verification) and the current study extends those findings in a sample of middle-aged adults (mean age = 33.9; Table 1) with no eligibility requirements for varying levels of substance use. As noted previously, this tripartite model is but one way of conceptualizing the interrelationships among self-regulatory traits and is almost certainly partially influenced by method variance (i.e., impulsive personality traits are measured using self-report questionnaires, impulsive choice is measured using showed preference decision making tasks, and impulsive action is measures using performance-based tasks). Nonetheless, the current results provide further evidence of its heuristic utility.

These findings together further support the idea that impulsivity is a multidimensional construct, one with distinct patterns of phenotypic and genetic relationships among indicators and with external phenotypes. Furthermore, though these results have shown a substantial proportion of impulsivity heritability can be recovered by PRSs, they cannot be used to delineate the origin of the shared genetic signals without a better understanding of the genetic architecture of the impulsivity phenotypes. Indeed, the reliable identification of individual SNPs will only be achievable in samples an order of magnitude or larger than this one. Discovery efforts remain crucial as the primary step towards closing the gap between SNP associations and heritability estimates, as well as between heritability estimates and polygenic prediction. Thus, larger and perhaps more environmentally homogenous study samples are necessary, as well as large and diverse ancestral samples as they become available.

Investigations on the genomic basis of impulsivity so far have been on variants with MAF >0.01. Research efforts towards mapping rare variants have already been underway for a variety of complex traits[49] and substance use phenotypes,[50] with promising results. Naturally, other forms of genetic and epigenetic variations might also contribute to or modulate impulsivity. For example, telomere length as a marker for biological aging has been linked to delay discounting, and accelerated aging was associated with synergistically steeper delay discounting in females compared with males.[51] Another line of research points to DNA-methylation patterns as a potential contributor, but is at an early stage.[52]

Collectively, the current investigation further informs the understanding of the genomic basis of impulsivity, confirming the multidimensionality of the phenotype, the constraints on detecting SNP-level associations in moderate sample sizes, and the promise of a PRS-based approach, in general and in samples of this size. As the genetic architecture of impulsivity gradually comes into clearer focus, so too can we expect to gain novel insights into the molecular, cellular, and system level determinants of self-regulatory traits and the associated health conditions.

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CONFLICT OF INTEREST STATEMENT

James MacKillop is a principal in Beam Diagnostics, Inc. and a consultant to Clairvoyant Therapeutics, Inc. No other authors have disclosures.

DATA AVAILABILITY STATEMENT

The summary statistics used to construct polygenic risk scores are available online with links provided in Table S3. The GTEx data used for the analyses described in this manuscript were obtained from: https://storage.googleapis.com/gtex_analysis_v8/single_tissue_qtl_data/GTEx_Analysis_v8_eQTL.tar, via the GTEx Portal on 10/09/2022. Summary statistics generated in the current study, including a total of 13 GWASs (three impulsivity domains and 10 individual impulsivity measures), will be made available on GWAS catalog.

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